

EXPERIMENT – 8

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Privilege Escalation Using TryHackMe Platform

Aim

To understand and demonstrate Linux privilege escalation techniques by exploiting system misconfigurations and vulnerabilities using the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Linux Privilege Escalation Tools (LinPEAS, sudo, find, etc.)

Theory (Brief)

Privilege Escalation is a technique used to gain higher-level permissions than originally assigned.

In Linux systems, attackers often exploit misconfigured permissions, vulnerable binaries, or weak sudo rules to escalate from a normal user to root.

TryHackMe provides a legal and controlled environment to practice privilege escalation techniques and understand real-world attack scenarios.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Privilege Escalation Room

1. In the search bar, type **Linux Privilege Escalation**.
2. Click on the appropriate room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Initial Enumeration

1. Check current user:

2. whoami
3. Check system information:
4. uname -a
5. Check sudo permissions:
6. sudo -l

Step 5: Identify Privilege Escalation Vectors

Common techniques explored:

- Sudo misconfigurations
- SUID binaries
- Weak file permissions
- Environment variable abuse
- Cron job misconfigurations

Example command:

```
find / -perm -4000 2>/dev/null
```

Step 6: Exploit Vulnerability

1. Execute the vulnerable binary or misconfigured command.
2. Escalate privileges to root user.
3. Verify escalation:
4. whoami

Step 7: Capture Flags / Complete Tasks

1. Obtain root-level access.
2. Capture the required flags.
3. Mark all tasks as completed.

Output

```
root@ip-10-49-191-120:~# whoami
root
root@ip-10-49-191-120:~# uname -a
Linux ip-10-49-191-120 6.17.0-1012-aws #12~24.04.1-Ubuntu SMP Mon Apr 6 17:36:28 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
root@ip-10-49-191-120:~# sudo -l
Matching Defaults entries for root on ip-10-49-191-120:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User root may run the following commands on ip-10-49-191-120:
    (ALL : ALL) ALL
root@ip-10-49-191-120:~# find / -perm -4000 -type f 2>/dev/null
/usr/lib/polkit-1/polkit-agent-helper-1
/usr/lib/x86_64-linux-gnu/dcv/dcvpamhelper
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/lib/xorg/Xorg.wrap
/usr/bin/passwd
/usr/bin/pkexec
/usr/bin/umount
/usr/bin/fusermount3
```

```
Applications Places System Sun Aug 30, 17:51 AttackBox IP:#
root@ip-10-49-191-120: ~
File Edit View Search Terminal Help

User root may run the following commands on ip-10-49-191-120:
(ALL : ALL) ALL
root@ip-10-49-191-120:~# find / -perm -4000 -type f 2>/dev/null
/usr/lib/polkit-1/polkit-agent-helper-1
/usr/lib/x86_64-linux-gnu/dcv/dcvpamhelper
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/lib/xorg/Xorg.wrap
/usr/bin/passwd
/usr/bin/pkexec
/usr/bin/umount
/usr/bin/fusermount3
/usr/bin/mount
/usr/bin/gpasswd
/usr/bin/su
/usr/bin/sudo
/usr/bin/chsh
/usr/bin/chfn
/usr/bin/newgrp
/usr/sbin/mount.nfs
/usr/sbin/pppd
/opt/BurpSuiteCommunity/burpbrowser/145.0.7632.45/chrome-sandbox
```

```
root@ip-10-49-191-120:~# cat /etc/crontab
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab'
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
# You can also override PATH, but by default, newer versions inherit
# it from the environment
#PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

# Example of job definition:
# .----- minute (0 - 59)
# | .----- hour (0 - 23)
# | | .----- day of month (1 - 31)
# | | | .----- month (1 - 12) OR jan,feb,mar,apr ...
# | | | | .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# | | | | |
# * * * * * user-name command to be executed
17 * * * * root cd / && run-parts --report /etc/cron.hourly
25 6 * * * root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.daily; }
```



THM AttackBox

Persistence: Off 49min 33s

root@ip-10-49-191-120: ~

File Edit View Search Terminal Help

```
# Example of job definition:
# .----- minute (0 - 59)
# | .----- hour (0 - 23)
# | | .----- day of month (1 - 31)
# | | | .----- month (1 - 12) OR jan,feb,mar,apr ...
# | | | | .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# | | | | |
# * * * * * user-name command to be executed
17 * * * * root cd / && run-parts --report /etc/cron.hourly
25 6 * * * root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.daily; }
47 6 * * 7 root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.weekly; }
52 6 1 * * root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.monthly; }
#
root@ip-10-49-191-120:~# getcap -r / 2>/dev/null
/usr/lib/snapd/snap-confine cap_chown,cap_dac_override,cap_dac_read_search,cap_fowner,cap_setgid,cap_setuid,cap_sys_chroot,cap_sys_ptrace,cap_sys_admin,cap_sys_resource=p
/usr/lib/x86_64-linux-gnu/gstreamer1.0/gstreamer-1.0/gst-ptp-helper cap_net_bind_service,cap_net_admin,cap_sys_nice=ep
```



THM AttackBox

Persistence: Off 49min 2s

```
Applications Places System Sun Aug 30, 17:54 AttackBox IP:#
root@ip-10-49-191-120: ~
File Edit View Search Terminal Tabs Help
root@ip-10-49-191-120: ~ x root@ip-10-49-191-120: ~ x
drwxr-xr-x 3 root root 4096 Mar 18 20:14 .shiv
drwx----- 2 root root 4096 Mar 11 09:12 .ssh
drwxr-xr-x 2 root root 4096 Feb 27 2026 .sstimap
drwxr-xr-x 2 root root 4096 May 14 06:02 .themes
drwx----- 6 root root 4096 May 15 12:22 .thunderbird
-rw----- 1 root root 10068 Jun 29 14:35 .viminfo
drwxr-xr-x 2 root root 4096 Apr 17 12:42 .vnc
-rw-r--r-- 1 root root 366 May 15 12:03 .wget-hsts
drwxr-xr-x 3 root root 4096 Apr 23 16:41 .wpscan
drwxr-xr-x 2 root root 4096 May 6 2024 CTFBuilder
drwxr-xr-x 4 root root 4096 May 26 13:15 Desktop
drwxr-xr-x 2 root root 4096 May 7 09:51 Documents
drwxr-xr-x 2 root root 4096 Feb 25 2026 Downloads
drwxr-xr-x 2 root root 4096 Feb 26 2026 Pictures
drwxr-xr-x 3 root root 4096 Feb 26 2026 Postman
drwxr-xr-x 47 root root 4096 Aug 13 10:42 Rooms
drwxr-xr-x 4 root root 4096 Aug 26 11:52 Scripts
drwxr-xr-x 2 root root 4096 May 7 09:39 Templates
drwxr-xr-x 3 root root 4096 Mar 18 10:48 go
drwxr-xr-x 2 root root 4096 Apr 23 19:06 neoreg_servers
drwxr-xr-x 3 root root 4096 May 15 06:40 reports
drwx----- 8 root root 4096 May 15 12:13 snap
drwxr-xr-x 2 root root 4096 Aug 25 11:53 test-reports
root@ip-10-49-191-120:~# x
THM AttackBox Persistence: Off 47min 19s
```

```
Applications Places System Sun Aug 30, 17:54
root@ip-10-49-191-120: ~
File Edit View Search Terminal Tabs Help
root@ip-10-49-191-120: ~ x root@ip-10-49-191-120: ~
root@ip-10-49-191-120:~# ls -la /root
total 224
drwx----- 43 root root 4096 Aug 25 11:48 .
drwxr-xr-x 23 root root 4096 Aug 30 17:42 ..
drwx----- 9 root root 4096 Jun 3 06:29 .BurpSuite
-rw----- 1 root root 1412 Aug 3 13:06 .ICEauthority
-rw----- 1 root root 1197 May 15 11:57 .Xauthority
drwxr-xr-x 21 root root 4096 Feb 25 2026 .ZAP
drwxr-xr-x 5 root root 4096 May 14 06:21 .attackbox-menu-back
260514-062154
lrwxrwxrwx 1 root root 9 Feb 27 2026 .bash_history -> /d
-rw-r--r-- 1 root root 3779 Jul 1 10:15 .bashrc
-rw-r--r-- 1 root root 3779 Jul 1 10:37
drwx----- 22 root root 4096 Aug 3 13:06 .cache
drwxr-xr-x 8 root root 4096 Mar 18 20:14 .cme
drwx----- 41 root root 4096 Aug 25 11:35 .config
drwx----- 3 root root 4096 Mar 3 09:07 .dbus
drwx----- 3 root root 4096 Feb 25 2026 .docker
drwx----- 3 root root 4096 Jul 15 16:42 .gnupg
drwx----- 2 root root 4096 Mar 3 09:07 .gvfs
drwxr-xr-x 2 root root 4096 May 14 06:02 .icons
drwxr-xr-x 4 root root 4096 Feb 25 2026 .java
drwx----- 3 root root 4096 May 15 12:14 .launchpadlib
```

Result

Thus, privilege escalation was successfully performed using the TryHackMe platform. The experiment provided hands-on understanding of Linux privilege escalation techniques and system security weaknesses.

⚠ Important Note

This experiment is performed **strictly for educational and ethical purposes** using the TryHackMe learning platform.